MATD12 – REPRESENTATION THEORY

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1. Lecture 1

1.1. **Setting the scene.** In many practical situations where we invoke groups in mathematics, it is not the group that is of importance but rather how it acts on other objects. Representation theory aims to ask the question: "What are all **things** that a given group can act on?" There are a few possible candidates for **things**. We will focus on vector spaces. Fix a field k.

Given a group G and a vector space V, we say that a homomorphism $\rho : G \to \mathrm{GL}(V)$ is a *representation* of G in V. The *degree* of ρ is defined to be $\dim(V)$ which may or may not be infinite.

An important remark is that we will often fix a basis of V and consider ρ as a homomorphism from G to the matrix group $GL_n(k)$ where n is the dimension of V. We will use these definitions interchangeably and without prior warning.

Whenever one defines an object in mathematics, one is morally obligated to also introduce a notion of morphisms. Given two representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$ we say that a linear map $\varphi : V \to W$ is a *morphism of representations* if for each $g \in G$ the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "\rho_g"] \arrow[d, "\varphi"'] & V \arrow[d, "\varphi"] \\ W \arrow[r, "\rho'_g"'] & W \end{tikzcd}$$

In particular, if φ is bijective, we shall say that ρ and ρ' are *isomorphic* representations.

Example 1.1. We will now give a non-example to show that an isomorphism of representations is stronger than an isomorphism of the underlying vector space. Consider the two representations on $\mathbb{C}$ of $C_2 = \langle x \mid x^2 = 1 \rangle$ given by

$$\rho_1 : x \mapsto 1 \qquad \text{and} \qquad \rho_2 : x \mapsto -1.$$

Notice that the underlying vector spaces are the same, but suppose for contradiction that there exists an isomorphism of representation $\varphi : \mathbb{C} \to \mathbb{C}$. There exists a nonzero $\lambda \in \mathbb{C}$ such that $\varphi = \lambda I$. Since φ was assumed to be a G-homomorphism it must be the case that

$$\begin{gathered} \rho_1(x)(\varphi(1)) = \varphi(\rho_2(x)1) \iff \\ \lambda = -\lambda \end{gathered}$$

which is a contradiction. Thus, these two representations are not isomorphic.

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1.2. **Representations as modules.** For the reader that knows what a module is, we see that the definition of a morphism above is indeed a natural one. Notice that given a group G we can form the vector space $k[G]$ for which the basis consists of all elements of G. We can define a multiplication on $k[G]$ by extending the group multiplication linearly to make $k[G]$ into an algebra. Then, we see that we can identify representations with $k[G]$-modules. Given a representation $\rho : G \to \mathrm{GL}(V)$ we give V the structure of a $k[G]$-module by imposing that

$$g \cdot v = \rho_g(v).$$

Similarly, given a $k[G]$-module V, we define a homomorphism

$$\begin{aligned} \rho : G &\longrightarrow \mathrm{GL}(V) \\ g &\longmapsto (v \mapsto g \cdot v). \end{aligned}$$

Under this identification, a morphism of representations is simply a homomorphism of $k[G]$-modules.

This perspective of representations as $k[G]$-modules will be important later on.

1.3. **First examples and further definitions.**

Example 1.2. Let us look at some first few examples.

(a) The map $\rho : \mathbb{R} \to \mathbb{R}^\times$ given by $\rho_t = e^t$ is a 1-dimensional representation of $\mathbb{R}$.
(b) The map $\rho : \mathbb{R} \to \mathrm{GL}(C(\mathbb{R}))$ given by $\rho_t(f)(x) = f(x + t)$ is an infinite-dimensional representation of $\mathbb{R}$.
(c) A representation of $\mathbb{Z}$ is simply an invertible linear operator $T : V \to V$.
(d) A rperesentation of $\mathbb{Z}/n\mathbb{Z}$ is a linear operator $T : V \to V$ such that $T^n = \mathrm{id}$.

Given a representation $\rho : G \to \mathrm{GL}(V)$ and a subspace $W \subseteq V$ for which $\rho_g(W) \subseteq W$ for all $g \in G$ we say that $\rho\,|_W : G \to \mathrm{GL}(W)$ is a subrepresentation of V. From the module point of view a subrepresentation is simply a $k[G]$-submodule.

Exercise 1.3. Prove that $\rho_g(W) \subseteq W$ for all $g \in G$ is equivalent to $\rho_g(W) = W$ for all $g \in G$. If we fix $g \in G$, is it true that $\rho_g(W) \subseteq W$ if and only if $\rho_G(W) = W$? What if W is finite-dimensional?

A nonzero representation $\rho : G \to \mathrm{GL}(V)$ is called *irreducible* if it has no proper nonzero subrepresentations.

Example 1.4. Consider the following examples.

(a) We can now ask whether the representation $\rho : \mathbb{R} \to \mathrm{GL}(C(\mathbb{R}))$ given by $\rho_t(f)(x) = f(x + t)$ is irreducible. The answer is obviously no. For instance, the subspace of 1-periodic functions is a subrepresentation. Let us determine all the 1-dimensional subrepresentations of ρ. If $f(x) \in C(\mathbb{R})$ generates a 1-dimensional subrepresentation, then there exist constants $\lambda(t)$ for each $t \in G$ such that $f(x+t) = \lambda(t) \cdot f(x)$ and so in particular $f(t) = \lambda(t) \cdot f(0)$. If $f(0) = 0$ then $f \equiv 0$ and thus does not generate a 1-dimensional subspace. If, on the other hand, $f(0) \neq 0$, then $\lambda(t) = f(t)/f(0)$ is continuous and satisfies that $\lambda(t + t') = \lambda(t)\lambda(t')$. One of the later exercises implies that $f(t) = f(0) \cdot e^{at}$ for some $a \in \mathbb{R}$. It is easy to check that all functions on this form do indeed generate a 1-dimensional subrepresentation and our characterisation is therefore completed.

(b) Which representations $\rho : \mathbb{Z} \to \mathrm{GL}_n(\mathbb{C})$ are irreducible? If $n = 1$, then ρ is certainly irreducible. We claim that if $n > 1$, then ρ cannot be irreducible. Indeed, we recall from linear algebra that ρ_1 will have an eigenvalue.

(c) Consider the representation $\rho : S_3 \to \mathrm{GL}_2(\mathbb{R})$ which is given by letting S_3 act on the vertices of a regular triangle in the plane. Clearly this action does not preserve any lines through the origin and is hence irreducible.

Consider now the complexification $\rho^{\mathbb{C}} : S_3 \to \mathrm{GL}_2(\mathbb{C})$. A more interesting question is whether this representation is irreducible. Suppose for the sake of contradiction that there exists a S_3-invariant 1-dimensional subspace $\langle v \rangle \subseteq \mathbb{C}^2$. Let $r \in S_3$ correspond to a rotation and $s \in S_3$ correspond to a reflection. Since $\rho_r^3 = \mathrm{id} = \rho_s^2$ we get that both ρ_r and ρ_s are diagonalisable and since $\rho_r \neq \mathrm{id}$ we deduce that there exists a third root of unity ϵ such that $\rho_r(v) = \epsilon v$ and $\rho_s(v) = \pm v$. This gives us that $\rho_{rs}(v) = \pm \epsilon v$, but rs corresponds to a reflection, so $\pm 1 v = \pm \epsilon v$. Thus $v = 0$, a contradiction. Hence $\rho^{\mathbb{C}}$ is irreducible.

Exercise 1.5. Prove that any continuous 1-dimensional representation of $\mathbb{R}$ is given by $f(t) = e^{at}$ for some $a \in \mathbb{R}$.

Exercise 1.6. Let us expand on an example above.

(a) Prove that any linear operator $A : V \to V$ where $\dim_{\mathbb{C}}(V) > 1$ has an eigenvalue.

(b) Show that the representation $\rho : \mathbb{Z} \to \mathrm{GL}_2(\mathbb{R})$ given by
$$\rho_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
is irreducible.

(c) Tweak the proof above, to show that any representation $\rho : \mathbb{Z} \to \mathrm{GL}_3(\mathbb{R})$ is reducible.

(d) We can do better. Show that for any $n > 2$, a representation $\rho : \mathbb{Z} \to \mathrm{GL}_n(\mathbb{R})$ is reducible.

1.4. **Maschke's theorem and indecomposable representations.** Our goal is to eventually be able to understand all representations of a given group (for sufficiently nice groups and over sufficiently nice fields). In a perfect world, we would be able to say that the study of irreducible representations lets us understand arbitrary representations completely. It turns out that we live in an (almost) perfect world.

Example 1.7. For a more serious example, we will consider the real representations of the dihedral group
$$D_8 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle.$$
We first find all the 1-dimensional representations. Suppose that $\phi : D_8 \to \mathbb{R}^\times$ is a representation. From the relations above one deduces that $\phi(r)^2 = \phi(s)^2 = 1$. Conversely each choice of $\phi(r)$ and $\phi(s)$ as ± 1 gives a representation of degree 1.

We also notice that there is degree 2 representation coming from the action of D_8 on the square situated in the plane. This representation $\rho : D_8 \to \mathrm{GL}(\mathbb{R}^2)$ can be given by
$$\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$
Can we find any more representations of D_8? The answer is yes for a silly reason. Think of the action of D_8 on a square situated in the xy-plane of $\mathbb{R}^3$. Then this new representation

can be given by

$$\rho'(r) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \rho'(s) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Repeating this trick, we can construct arbitrarily large representations.

In the last example, we saw that we can "glue together" representations. This motivates the following definition. Given two representations $\rho_1 : G \to \mathrm{GL}(V)$ and $\rho_2 : G \to \mathrm{GL}(W)$ we can form their *direct sum* $\rho_1 \oplus \rho_2 : G \to \mathrm{GL}(\mathrm{V} \oplus \mathrm{W})$ defined by

$$\rho(s)(v, w) = (\rho_1(s)v, \rho_2(s)w).$$

If ρ is a representation of G such that there do not exist proper subrepresentations ρ_1 and ρ_2 with $\rho_1 \oplus \rho_2 \cong \rho$ we say that ρ is *indecomposable*.

Exercise 1.8. Show that any irreducible representation is indecomposable.

Example 1.9. Let us consider two cases where the notion of irreducible and indecomposable representations differ.

(a) Consider the representation of $C_2 = \langle x \mid x^2 = 1 \rangle$ over the field $\mathbb{F}_2$ given by $\rho : C_2 \to \mathrm{GL}(\mathbb{F}_2^2)$ and

$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

This is a representation since

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is **not irreducible** since the subspace generated by $(1, 0)$ gives a subrepresentation. However, one can show that this is the only subrepresentation of ρ and thus ρ is **indecomposable**.

(b) Consider the related representation $\rho : \mathbb{Z} \to \mathrm{GL}_2(\mathbb{C})$ given by

$$\rho_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Similarly to above, one can show that ρ_1 is not irreducible, yet indecomposable.

This is troubling, since understanding all representations is equivalent to understanding the indecomposable representations. On the other hand, irreducible representations will be easier for us to deal with. These two notions converge when G is finite and $\mathrm{char}(k) = 0$. Please note that we only use that $|G| \neq 0$ in k in our proof and hence it suffices to assume that $\mathrm{char}(k) \nmid |G|$.

Theorem 1.10 (Maschke)**.** *Let k be a field of characteristic 0 and G a finite group. If $\rho : G \to \mathrm{GL}(V)$ is indecomposable, then it is irreducible.*

We will first give a proof of a geometric flavour that only works when $k = \mathbb{R}$ or $\mathbb{C}$ and V is finite-dimensional.

Proof. We will prove the contrapositive. Suppose that $W \subseteq V$ is a proper nonzero subrepresentation. Since V is finite-dimensional over $\mathbb{R}$ or $\mathbb{C}$ it admits an inner product B. Using this inner-product we could define $W^\perp$ and then clearly $V = W \oplus W^\perp$ but the problem is that $W^\perp$ need not be a subrepresentation (i.e. G-invariant). To fix this, we will use an averaging trick. Consider the inner product $B' = \sum_{g \in G} B(\rho_g v, \rho_g w)$ which is well-defined since G is finite. Clearly $B'(\rho_g v, \rho_g w) = B'(v, w)$ for all $g \in G$ (the terms in the sum are simply permuted). Now, let $W^\perp$ be the orthogonal complement of W with respect to this inner product B'. Then, it follows that $W^\perp$ is G-invariant since for any $v \in W^\perp$ and any $w \in W$:

$$B'(\rho_g v, w) = B'(v, \rho_{g^{-1}} w) = 0$$

since W is G-invariant. Thus, $V = W \oplus W^\perp$ is a reducible representation. The theorem follows. □

Exercise 1.11. Deduce the following for any finite group G using our geometric proof of Maschke's theorem:

(a) If $\rho : G \to \mathrm{GL}_n(\mathbb{R})$ is a representation, then there exists a basis of $\mathbb{R}^n$ such that $\mathrm{Im}\rho \subseteq O_n$.
(b) If $\rho : G \to \mathrm{GL}_n(\mathbb{C})$ is a representation, then there exists a basis of $\mathbb{C}^n$ such that $\mathrm{Im}\rho \subseteq U_n$.

Exercise 1.12. Explain why any finite-dimensional representation of a finite group over a field of characteristic 0 can be decomposed as a direct sum of irreducible representations. Does the same result hold for infinite-dimensional representations? For infinite groups? For fields of positive characteristic?

2. Lecture 2

2.1. **A fun exercise.** Recall the complex representation

$$\rho : \mathbb{R} \longrightarrow \mathrm{GL}(C(R))$$
$$\rho_t(f)(x) = f(x+t).$$

Last time, we showed that all 1-dimensional subrepresentations are given by $\langle e^{ax} \rangle$. For 2-dimensional subrepresentations, we can for instance find $\langle e^{ax}, xe^{ax} \rangle$.

Exercise 2.1. Is it true that the only n-dimensional indecomposable subrepresentation is

$$\langle e^{ax}, xe^{ax}, x^2e^{ax}, \ldots, x^{n-1}e^{x} \rangle.$$

2.2. **A second proof of Maschke's theorem.** We will now give a proof for Mascke's theorem (Theorem 1.10) that works in general.

Proof. Let G be a finite group, k be a field of characteristic 0 and $\rho : G \to \mathrm{GL}(V)$ a representation. Suppose that $W \subseteq V$ is a subrepresentation.

From linear algebra, we know that there exists a projection $\pi' : V \to V$ such that $\pi'(V) \subseteq W$ and $\pi'|_W = \mathrm{id}_W$. We will first try to find a projection which is a morphism of representations. This can be done by an averaging trick. Consider the linear map $\pi : V \to V$ given by

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} \rho_g \circ \pi'(\rho_{g^{-1}} v).$$

This is clearly a projection and a morphism of representations from $V \to V$. You will show in the exercises that the category of representations is abelian so $W' = \ker(\pi)$ is a subrepresentation of V and since π is a projection $W \oplus W' \cong V$ which proves that V is decomposable. This finishes the proof. $\square$

Exercise 2.2. In this exercise, we will verify the computations in the above proof that were left out.

(a) Prove that if $W \subseteq V$ are vector spaces then there exists a projection $\pi' : V \to V$ such that $\pi'(V) \subseteq W$ and $\pi'|_W = \mathrm{id}_W$.
(b) Prove that π above is a projection.
(c) Prove that π is a morphism of representations.

Exercise 2.3. It turns out that the category of representations of a group G over a field k is abelian. You are not expected to know what this means. Instead, prove that if φ is a morphism of representations, then $\ker(\varphi)$ and $\mathrm{im}(\varphi)$ are representations.

2.3. **Schur's lemma.** In this section, our treatment will deviate somewhat from the lecture. The main theorem of this sections is the following.

Theorem 2.4 (Schur)**.** *Let (V, ρ) and (V', ρ') be representations of a group G over a field k and let $\varphi : V \to V'$ be a morphism of representations. Then:*

(i) if V is irreducible, then φ is injective.
(ii) if V' is irreducible, then φ is subjective.
(iii) if $(V, \rho) = (V', \rho')$ is irreducible and finite dimensional and k is algebraically closed, then $\varphi = \lambda I$ for some $\lambda \in k^\times$.

Proof. Part (i) follows from the fact that $\ker(\varphi)$ is a subrepresentation of V and part (ii) follows from the fact that $\mathrm{im}(\varphi)$ is a subrepresentation of V' using Exercise 2.3.

For part (iii), assume that $\varphi : V \to V$ is a nonzero endomorphism of an irreducible representation over an algebraically closed field k. Since k is algebraically closed, there exists an eigenvalue $\lambda \in k$ of φ. In particular, $\varphi - \lambda \cdot \mathrm{id} : V \to V$ is a morphism of V which is not injective. Hence, by part (i), $\varphi - \lambda \cdot \mathrm{id} = 0$ which finishes the proof. $\square$

What is the upshot of Schur's lemma? Let V and W be representations of a group G over a fixed base field k. We will let $\mathrm{Hom}_G(V, W)$ denote the vector space of G-homomorphisms from V to W and $\mathrm{End}_G(V)$ the $k[G]$-algebra of G-endomorphisms of V. Then:

- If V and W are nonisomorphic representations, then $\dim \mathrm{Hom}_G(V, W) = 0$.
- If $V \cong W$, then $\dim(\mathrm{Hom}_G(V, W)) > 0$.
- If V is irreducible, then the k-algebra $\mathrm{End}_G(V)$ is a division algebra. This means that every nonzero element is invertible. If k is algebraically closed, then $\mathrm{End}_G(V) \cong k$.

In the next few lectures, we will see that the very successful theory of characters is essentially a corollary of Schur's lemma. In particular, we will see that $\dim(\mathrm{Hom}_G(-, -))$ gives us a way to take an inner product of representations.

Example 2.5. As a last example, we will observe that if k is not algebraically closed, then $\mathrm{End}_G(V)$ need not be a 1-dimensional vector space over k, even if V is irreducible.

Consider the representation (V, ρ) given by

$$\begin{aligned} \rho : C_3 &\longrightarrow \mathrm{GL}_2(\mathbb{R}) \\ 1 &\longmapsto A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \end{aligned}$$

which is a representation over $\mathbb{R}$ since $A^3 = \mathrm{id}$. This representation is irreducible since the eigenvalues of A over $\mathbb{C}$ are nonreal. What does $\mathrm{End}_{C_3}(V)$ look like? It is the collection of matrices $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ such that $BA = AB$, which are exactly the matrices on the form

$$\begin{pmatrix} a & -b \\ b & a-b \end{pmatrix} = a \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \cdot \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

Thus,

$$\mathrm{End}_{C_3}(V) = \left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \right\rangle$$

has dimension 2. As a $\mathbb{R}$-algebra,

$$\mathrm{End}_{C_r}(V) \cong \mathbb{R}[e^{2\pi i/3}].$$

What is going on here? Recall that this representation is equivalent to a $\mathbb{R}[G]$-module V. When we tensor this with $\mathbb{C}$ we get a $\mathbb{C}[G]$-module $\mathbb{C} \otimes V$ which is decomposable as the direct sum of the representations $1 \mapsto e^{2\pi i/3}$ and $1 \mapsto e^{4\pi i/3}$. The lack of these two elements in the field $\mathbb{R}$ is what is making the situation more complicated.

Exercise 2.6. Let n be a positive integer.

(a) What are the irreducible representations of $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{C}$?
(b) Let p be a prime. What are the irreducible representations of $\mathbb{Z}/p\mathbb{Z}$ over $\mathbb{Q}$?
(c) What are the irreducible representations of $\mathbb{Z}/4\mathbb{Z}$ over $\mathbb{Q}$?

3. Lecture 3

In this lecture, we will explore applications of Schur's lemma. This forms the backbone of the theory of representations of finite groups over algebraically closed fields.

3.1. Representations of abelian groups.

Theorem 3.1. *Let G be an abelian group and k be an algebraically closed field. Then, every irreducible representation of G over k is 1-dimensional.*

Proof. Let $\rho : G \to \mathrm{GL}(V)$ be an irreducible representation. Since G is abelian we can check that for every $g \in G$, ρ_g is a morphism of representations $V \to V$. By Schur's lemma, $\rho_g = \lambda_g I$ for some scalar $\lambda_g \in k^\times$. Since any vector space is invariant under multiplication by scalars, it follows that if V is irreducible, then $\dim(V) = 1$. $\square$

3.2. **General constructions.** Given two representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(V)$ of a group G over a field k, we already know one way to combine them, namely to take the direct sum. Unfortunately, this does not really give us any additional information about G and its representations. However, there are more fruitful things we can do.

For instance, we can form the *tensor product* $\rho \otimes \rho' : G \to \mathrm{GL}(V \otimes_k V')$ which is defined by $(\rho \otimes \rho')_g(v \otimes v') = \rho_g(v) \otimes \rho'_g(v')$. As a $k[G]$-module, this is of course simply $V \otimes_k W$.

We can consider a new representation $\psi : G \to \mathrm{GL}(\mathrm{Hom}(V, V'))$ given by

$$\psi_g(f) = \rho'_g \circ f \circ \rho_{g^{-1}}.$$

Lastly, if we have any group action of G on a finite set X, we can define an representation of G in $k[X]$ by:

$$\rho_h(\sum_g a_g \cdot g) = \sum_g a_{h^{-1}g} \cdot g.$$

The most interesting example is when $X = g$, in which case we get the *regular representation* $k[G]$ which is by far the most important representation as we shall see in the following section.

3.3. **Classification of irreducible representations.** Let $\rho : G \to \mathrm{GL}(V)$ be a representation. We leave it as a straightforward exercise to show the following important lemma.

Lemma 3.2. *We have the following isomorphism of vector spaces:*

$$\begin{aligned} \mathrm{Hom}_G(k[G], V) &\xrightarrow{\sim} V, \\ f &\longmapsto f(1_G). \end{aligned}$$

Notice that this is an isomorphism on the level of vector spaces and not between representations.

We now state our most important theorem.

Theorem 3.3. *Let G be a finite group and k a field of characteristic 0.*

(i) Every irreducible representation of G is isomorphically a subrepresentation of $k[G]$.
(ii) There are only finitely many irreducible representations of G up to isomorphism.
(iii) If k is algebraically closed and $P_1, \dots, P_m$ are all irreducible representations of G, then

$$\sum \dim(P_i)^2 = |G|.$$

Proof. By Maschke's theorem we can write

$$k[G] \cong \oplus_i P_i^{n_i}$$

for pairwise nonisomorphic irreducible representations P_i. By Lemma 3.2, for every irreducible representation V of G,

$$\begin{aligned} V &\cong \mathrm{Hom}_G(k[G], V) \\ &\cong \mathrm{Hom}_G(\oplus_i P_i^{n_i}, V) \\ &\cong \oplus_i n_i \mathrm{Hom}_G(P_i, V). \end{aligned}$$

By Schur's lemma, and $\dim(V) > 0$ we obtain that $P_i \cong V$ for some i. This proves (i) and (ii) follows immediately.

For (iii), note that $\dim(V) = n_i$ by Schur's lemma. Hence, (iii) follows from $\dim(k[G]) = |G|$. □

Example 3.4. We will now try to classify the irreducible representations for some symmetric groups over $\mathbb{C}$.

(a) Consider S_2. Since $2 = 1^2 + 1^2$ there are only two irreducible representations each of degree 1 and these are clearly the trivial representation and the sign representation.
(b) In general, let us find the 1-dimensional representations of S_n in two ways. One way is to notice that the abelianisation of G is $S_n/A_n \cong C_2$ and hence the only homomorphisms $G \to \mathbb{C}^\times$ are the trivial and sign map.

Another way to see this is to consider the generators $s_k = (1 \quad k) \in S_n$. Finding a 1-dimensional representation requires solving the equations

$$\begin{aligned} s_i^2 &= 1 && \text{for } 1 \le i \le n \\ (s_i s_j)^3 &= 1 && \text{for } 1 \le i < j \le n \end{aligned}$$

over $\mathbb{C}^\times$, which is easily done and gives exactly the trivial representation and the sign representation.

(c) Consider S_3. Since $3! = 1^2+1^2+2^2$ we see that besides the 1-dimensional irreducible representations, we have a unique irreducible representation of dimension 2 which we have already seen is the symmetries of a regular triangle in the plane.

(d) For a general symmetric group S_n we clearly have a representation of S_n in $\mathbb{C}[X]$ where $X = \{1, \dots, n\}$. Furthermore, we see that this representation is isomorphic to a direct sum of the representations:

$$V = \langle (1, 1, \dots, 1) \rangle,$$
$$W = \{(x_1, \dots, x_n) : x_1 + \cdots + x_n = 0\}.$$

We claim that W is irreducible. Suppose not and that we have a nonzero proper subrepresentation $U \subseteq W$. This must contain a vector $(x_1, \dots, x_n)$ with $x_1 \neq x_2$. Hence, it also contains

$$(x_1, x_2, x_3, \dots, x_n) - (x_2, x_1, x_3, \dots, x_n) = (x_1 - x_2, x_2 - x_1, 0, \dots, 0).$$

It follows that

$$(1, -1, 0, \dots, 0), (1, 0, -1, 0, \dots, 0), \dots, (1, 0, \dots, 0, -1) \in U.$$

These vectors span V and hence $U = V$, a contradiction. Thus W is irreducible.

(e) Lastly, consider S_4. We have two irreducible 1-dimensional representations. By part (d), we have a 3-dimensional representation ρ and tensoring with the sign representation we get a new representation ρ'. These are non-isomorphic since the traces

$$\mathrm{Tr}(\rho_{(1\,2)}) = 1 \qquad \mathrm{Tr}(\rho'_{(1\,2)}) = -1$$

differ.

Since $24 = 1^2 + 1^2 + 3^2 + 3^2 + 2^2$ we have left to find one 2-dimensional irreducible representation. Notice that $V_4 \trianglelefteq S_4$ and hence we have surjection $S_4 \to S_4/V_4 \cong S_3$. Hence, the irreducible representation of S_3 gives us the unique irreducible representation of S_4 for free and our classification is completed.

Exercise 3.5. Prove that if we have a surjective group homomorphism $\varphi : G \to H$ and an irreducible representation $\rho : H \to \mathrm{GL}(V)$, then $\rho \circ \varphi : G \to \mathrm{GL}(V)$ is an irreducible representation.

Exercise 3.6. Let k be an algebraically closed field of characteristic 0 and G be a finite group. Prove that the number of 1-dimensional characters of G is equal to $|G/[G, G]|$.

Exercise 3.7. Let K/k be a field extension and let $\rho : G \to \mathrm{GL}(V)$ be a representation over k.

(a) Prove that

$$\dim(\mathrm{End}_{k[G]}(V)) = \dim(\mathrm{End}_{K[G]}(K \otimes V)).$$

(b) Deduce that if k and K are algebraically closed, then V is irreducible over k if and only if the representation $K \otimes_k V$ is irreducible over K.

4. Lecture 4

In this lecture, we will assume that k is an algebraically closed field of characteristic 0 and that G is a finite group with irreducible representations $P_1, \dots, P_m$. Our main goal is to prove that m is equal to the number of conjugacy classes of G.

4.1. **Module point of view.** Let us recall some terminology that we introduced in the first lecture. Recall that $k[G]$ is a k-algebra called the *group algebra* and that a representation of G in V over k is the same as a homomorphism of k-algebras:

$$k[G] \longrightarrow \mathrm{End}_k(V).$$

4.2. **Describing the group algebra.**

Theorem 4.1. *The map $\varphi : k[G] \longrightarrow \mathrm{End}_k(P_1) \times \cdots \times \mathrm{End}_k(P_m)$ is an isomorphism of k-algebras.*

Proof. Let $z \in k[G]$ be such that $\varphi(z) = 0$. Then, by Maschke's theorem, it follows that z acts trivially on any finite dimensional representation of G. In particular, it acts trivially on $k[G]$. This implies that $z \cdot e = 0$ and hence $z = 0$. This shows that φ is injective.

In the last lecture, we proved the formula $|G| = \sum \dim(P_i)^2$. This shows that

$$\dim(k[G]) = \dim(\mathrm{End}_k(P_1) \times \cdots \times \mathrm{End}_k(P_m))$$

and hence φ is an isomorphism of k-algebras, which was what we wanted to show. $\square$

Example 4.2. Notice that the above property is special for *group algebras* and relies heavily on Maschke's theorem. In particular consider the k-algebra of *dual numbers* $k[\epsilon]/(\epsilon^2)$. Suppose that M is an irreducible module over this algebra and we have a map $\rho : k[\epsilon]/(\epsilon^2) \to \mathrm{End}_k(M)$. Note that $\ker \rho_\epsilon$ is stable and that ρ_ϵ is not injective. Thus, since M is irreducible, $\ker(\rho_\epsilon) = M$ and $M \cong k$. Thus, ϵ is a nonzero element of the group algebra acting trivially on all irreducible modules.

We can now deduce the main theorem of this lecture.

Theorem 4.3. *The number of isomorphism classes of irreducible representations of G over k is equal to the number of conjugacy classes of G.*

Proof. A quick computation shows that $\dim(Z(k[G]))$ is equal to the number of conjugacy classes in G. Additionally,

$$\dim(Z(\mathrm{End}_k(P_1)) \times \cdots \times Z(\mathrm{End}_k(P_m))) = \sum_{i=1}^{m} 1 = m$$

since the center of a matrix algebra is 1-dimensional. The isomorphism in Theorem 4.1 finishes the proof. $\square$

Exercise 4.4. Recall that we already showed using Schur's lemma that if G is abelian, then any irreducible representation has dimension 1. Prove the converse: if every irreducible representation of G has dimension 1, then G is abelian.

5. Lecture 5

In this lecture, we develop a computational tool for determining the irreducible representations of a finite group, namely *character theory*. Please note that there is not much new mathematical content in this lecture and that the theory of characters is essentially a useful corollary of the previous lecture. We will fix an algebraically closed field k of characteristic 0 and a finite group G. Furthermore, we will denote the irreducible representations of G over k by $P_1, \ldots, P_m$.

5.1. **Character theory.** Recall that we have an isomorphism of k-algebras:

$$k[G] \xrightarrow{\simeq} \prod_i \mathrm{End}(P_i).$$

In particular, this restricts to an isomorphism of vector spaces

$$A : Z(k[G]) \xrightarrow{\simeq} \prod_i k \cdot \mathrm{Id}.$$

Note that we have natural choices of bases on both the left and right side of the above isomorphism. Hence, we can try to understand concretely what the inverse of this map looks like. Additionally, recall that we can think of $Z(k[G])$ as functions on the conjugacy classes of G. Notice also that for any $\varphi \in \prod_i \mathrm{End}(P_i)$ can be interpreted as an endomorphism of $k[G]$, since

$$k[G] \cong P_1^{\dim(V_1)} \oplus \cdots \oplus P_m^{\dim(V_m)}.$$

Theorem 5.1. *The inverse of A is given by:*

$$B : \prod_i k \cdot \mathrm{Id} \longrightarrow Z(k[G]),$$

$$\varphi \longmapsto \frac{1}{|G|}\mathrm{Tr}(g^{-1} \circ \varphi : k[G] \to k[G]).$$

Proof. The proof follows from a computation. It is sufficient to verify that B is a left inverse of A. Indeed, for any $h \in G$:

$$\begin{aligned} B(A(h)) &= \frac{1}{|G|}\mathrm{Tr}(g^{-1} \circ h) \\ &= h \in k[G]. \end{aligned}$$

The latter equality follows from the fact that $g^{-1} \circ h$ permutes the natural basis elements of $k[G]$ and hence the trace above is $|G|$ when $g = h$ and 0 otherwise. $\square$

Please note that there is some slight abuse of notation above. Let us now explore where each basis vector

$$E_i = (0, \ldots, \mathrm{id}, \ldots, 0)$$

is sent by B. For this we will need a new definition. If $\rho : G \to \mathrm{GL}(V)$ is a finite dimensional representation, then we define its *character* $\chi_\rho \in k[G]$ by $\chi_\rho(g) = \mathrm{Tr}(\rho_g)$ which is well-defined since the trace is invariant under conjugation.

Now, we can note that:

$$\begin{aligned} B(E_i) &= \frac{1}{|G|}\mathrm{Tr}(g^{-1} \circ E_i) \\ &= \frac{\dim(P_i)}{|G|}\mathrm{Tr}(g^{-1} : P_i \to P_i) \\ &= \frac{\dim(P_i)}{|G|}\chi_{P_i}(g^{-1}). \end{aligned}$$

We can define a symmetric inner product on $k[G]$ by

$$(f, h) = \frac{1}{|G|}\sum_g f(g) \cdot h(g^{-1})$$

for $f, h \in G$.

Theorem 5.2. *We have the the following results:*

(i) $(\chi_{P_i}, \chi_P) = \delta_{ij}$,
(ii) $\rho : G \to \mathrm{GL}(V)$ *is irreducible if and only if* $(\chi_\rho, \chi_\rho) = 1$,
(iii) $\rho : G \to \mathrm{GL}(V)$ *and* $\rho' : G \to \mathrm{GL}(W)$ *are isomorphic if and only if* $\chi_\rho = \chi_{\rho'}$.

Proof. Obviously $E_i E_j = \delta_{ij} E_i$ in $\prod_i \mathrm{End}(P_k)$. In $k[G]$, this translates To

$$\left(\sum \chi_{\rho_i}(g^{-1})g\right)\left(\sum \chi_{\rho_j}(g^{-1})g\right) = \delta_{ij}\frac{|G|}{\dim(P_i)}\sum \chi_{\rho_i}(g^{-1})g.$$

Equating the coefficients of $g = 1$, we obtain that

$$\sum \chi_{\rho_i}(g)\chi_{\rho_j}(g^{-1})g = \delta_{ij}|G| \iff (\chi_{\rho_i}, \chi_{\rho_j}) = \delta_{ij},$$

which proves (i). Notice that if $\rho : G \to \mathrm{GL}(V)$, then there exist nonnegative integers n_i such that $V \cong \oplus_i P_i^{n_i}$. Then,

$$(\chi_\rho, \chi_\rho) = \sum_i n_i^2$$

is equal to 1 if and only if one n_i is 1 and the rest are 0. This proves (ii). Moreover,

$$(\chi_\rho, \chi_{P_i}) = n_i$$

which proves (iii). □

The next theorem shows that we can obtain a canonical projection of a representation $V \cong \oplus P_i^{n_i}$ onto $P_i^{n_i}$. This is referred to as a canonical decomposition into *isotypical components*. It is a straightforward exercise and will be left to the reader.

Theorem 5.3. *Consider*

$$\pi_i := \frac{\dim(P_i)}{|G|}\sum_g \chi_{\rho_i}(g^{-1})g \in k[G].$$

Then π *acts like the identity on* P_j *when* $i = j$ *and like* 0 *when* $i \neq j$. *Furthermore, for any representation* V,

$$\rho_{\pi_i} : V \to V$$

is a G*-homomorphism and a projection onto the direct sum of all the copies of* P_i *in* V.

Let us consider one of the simplest examples of the above theorem.

Example 5.4. Let $G = \mathbb{Z}/n\mathbb{Z}$. Let $\omega = e^{2\pi i/n}$ and consider the irreducible characters of G given by

$$\begin{aligned} \rho_j : G &\longrightarrow \mathbb{C}^\times \\ 1 &\longmapsto \omega^j. \end{aligned}$$

Then, given any representation $V = \oplus P_i^n$, we have the projections:

$$\rho_{\pi_i}(v) = \frac{1}{n}\sum_{\bar{m}\in\mathbb{Z}/n\mathbb{Z}} \omega^{-m}\cdot\rho_{\bar{m}}(v).$$

For example, when $n = 2$, we have the two projections

$$\rho_{\pi_0} = \frac{1}{2}(\rho_0 + \rho_1) \quad \text{and} \quad \frac{1}{2}(\rho_0 - \rho_1).$$

As a special case, if W is any vector space, then we have a representation of $\mathbb{Z}/2\mathbb{Z}$ on $W \otimes W$ given by

$$\rho_1(w_1 \otimes w_2) = w_2 \otimes w_1.$$

Using this, we get a decomposition

$$W \otimes W = S^2W \oplus \Lambda^2 W$$

where S^2W denotes the symmetric tensors and $\Lambda^2 W$ the alternating tensors.

5.2. **Character tables.** Given a group G, we can construct its character table. Two examples are given below.

Example 5.5. Let $G = S_3$. We have already constructed all the irreducible representations in earlier lectures and we therefore see that the we have the character table below.

	1	(1 2)	(1 2 3)
trivial	1	1	1
sign	1	-1	1
symmetries of a triangle	2	0	-1

We will soon proceed to a more complicated example, but we will first need a lemma.

Lemma 5.6. *Suppose that $H \leq G$ is an abelian subgroup. Then any irreducible representation of G has order $\leq \frac{|G|}{|H|}$.*

Proof. Let V be an irreducible representation of G. As a representation of H, it has a 1-dimensional subrepresentation W. Then, if $G = \cup_i g_i H$, we see that $\sum \rho_{g_i}(W) \subseteq V$ is a subrepresentation of G. Since V is irreducible we have that $V = \sum \rho_{g_i}(W)$ and hence $\dim(V) \leq |G|/|H|$ as claimed. $\square$

Example 5.7. Let $G = D_4$ and recall that $|D_4| = 8$. Since $C_4 \leq D_4$ all irreducible representations of D_4 are of dimension ≤ 2. To find the 1-dimensional representations of D_4, we simply have to solve the system of equations

$$\begin{aligned} r^4 &= s^2 = 1, \\ srs &= r^{-1}. \end{aligned}$$

which gives solutions $r, s \in \{\pm 1\}$. Thus, all 1-dimensional representations are real and since the 2-dimensional representation of D_4 as symmetries of a square is irreducible over $\mathbb{R}$, it is clear that it is also irreducible over $\mathbb{C}$ for this reason. Hence, we obtain the character table below.

e	$\{r, r^3\}$	$\{r^2\}$	$\{s, sr^2\}$	$\{sr, sr^3\}$
1	1	1	1	1
1	-1	1	1	-1
1	-1	1	-1	1
1	1	1	-1	-1
2	0	-2	0	0

5.3. **Constructing new representations.** One might be misled by character theory to think that the subject of the representation theory of finite groups is trivial. This is not the case. The hard part is to generate enough irreducible representations. One fruitful way we have seen to construct new representations is by taking hte tensor product and we shall see that the character of the tensor product is easy to compute.

Proposition 5.8. *If $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$ are representations, then the character of $\rho \otimes \rho' : G \to GL(V)$ is given By*

$$\chi_{\rho\otimes\rho'}(g) = \chi_\rho(g) \cdot \chi_{\rho'}(g).$$

The proof follows from the below lemma from linear algebra.

Lemma 5.9. *If V and W are finite-dimensional vector spaces and $A \in \mathrm{End}(V)$ and $B \in \mathrm{End}(W)$, then $\mathrm{Tr}(A \otimes B) = \mathrm{Tr}(A) \cdot \mathrm{Tr}(B)$.*

Proof. To prove this, note that both the left and right hand side are linear in A and in B. Thus, it is sufficient to prove this claim when $A = E_{ij}$ and $B = E'_{i'j'}$ for some elementary matrices. This is an easy computation. $\square$

6. Lecture 6

6.1. **Constructing new representations.** Last lecture, we discussed how to obtain new representations from old by taking the tensor product. Today, we will give a couple of further examples. Given a representation $\rho : G \to \mathrm{GL}(V)$, we define the *dual representation* $\rho^* : G \to \mathrm{GL}(V^*)$ by

$$(\rho_g^*(f))(v) \coloneqq f(\rho_g^{-1}(v))$$

for each $f \in V^*$, $v \in V$ and $g \in G$. If we choose a basis for V and its dual basis for V^*, we obtain a map

$$\begin{aligned} \mathrm{GL}_n(k) &\longrightarrow \mathrm{GL}_n(k) \\ A &\longmapsto (A^t)^{-1} \end{aligned}$$

such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\rho"] \arrow[dr, "\rho^*"'] & \mathrm{GL}_n(k) \arrow[d] \\ & \mathrm{GL}_n(k) \end{tikzcd}$$

Hence, we see that the character of ρ^* is

$$\chi_{\rho^*}(g) = \chi_\rho(g^{-1})$$

for $g \in G$.

Moving forward we let $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$ be representations. Notice that $\mathrm{Hom}_k(V, W) \cong V^* \otimes W$ when V is finite-dimensional. Using that $V^* \otimes W$ is a representation of G, we get that $\mathrm{Hom}_k(V, W)$ is a representation of G and that it is given By

$$\rho'' : G \to \mathrm{Hom}_k(V, W)$$
$$\rho''_g(f)(v) = \rho'_g(f(\rho_g^{-1}(v))).$$

Note that we can define $V^G = \{v \in V : \rho_g(v) = v \text{ for all } g \in G\} \subseteq V$ and that G acts trivially on this set.

Exercise 6.1. Show that $\mathrm{Hom}_k(V,W)^G = \mathrm{Hom}_G(V,W)$.

We can now show a theorem that generalises the orthogonality relations for irreducible representations over an algebraically closed field of characteristic 0.

Theorem 6.2. *Let k be a field such that $\mathrm{char}(k) \nmid |G|$ and consider two finite-dimensional representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$. Then*

$$(\chi_\rho, \chi_{\rho'}) = \dim_k \mathrm{Hom}_G(W,V).$$

Proof. We first remark that that for any representation $\rho'' : G \to \mathrm{GL}(Q)$, the map

$$\pi = \frac{1}{|G|}\sum_{g\in G} \rho''_g$$

is a projection onto Q^G. Thus,

$$\begin{aligned}(\chi_\rho, \chi_{\rho'}) &= \frac{1}{|G|}\sum_{g\in G} \chi_\rho(g)\cdot \chi_{\rho'}(g^{-1})\\ &= \frac{1}{|G|}\sum_{g} \chi_{W^*\otimes V}(g)\\ &= \mathrm{Tr}\left(\frac{1}{|G|}\sum_{g\in G} \rho_{W^*\otimes V}(g)\right)\\ &= \dim((W^*\otimes V)^G)\\ &= \dim(\mathrm{Hom}_G(W,V))\end{aligned}$$

using the previous remark. □

Exercise 6.3. Verify some of the linear algebra facts we used in the proof of Theorem 6.2.

(a) Prove that π in the theorem is a projection.
(b) Prove that for any linear projection $\pi : V \to V$, we have that $\mathrm{Tr}(\pi) = \dim(\pi(V))$.

Exercise 6.4. Prove that if $k = \mathbb{C}$, then $\chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$.

6.2. **Representations of products.** In this subsection, we consider a field $\mathbb{Q} \subseteq k = \bar{k}$. We shall show that to understand the representations of a product of groups, it is sufficient to understand the representations of the components. Note first that if G_1 acts on V_1 and G_2 acts on V_2, then $G_1 \times G_2$ acts naturally on $V_1 \otimes_k V_2$.

Proposition 6.5. *If $P_1,\dots,P_d$ are the irreducible representations of G_1 and $Q_1,\dots,Q_m$ are the irreducible representations of G_2, then all irreducible representations of $G_1\times G_2$ are given by $P_i \otimes Q_j$.*

Proof. Notice firstly that

$$\begin{aligned}(\chi_{P_i\otimes Q_j}, \chi_{P_{i'}\otimes Q_{j'}}) &= \frac{1}{|G_1|\cdot|G_2|}\sum_{g_1\in G_1,\, g_2\in G_2} \chi_{P_i}(g_1)\cdot\chi_{Q_j}(g_2)\cdot\chi_{P_{i'}}(g_1^{-1})\cdot\chi_{Q_{j'}}(g_2^{-1})\\ &= \left(\frac{1}{|G_1|}\sum_{g\in G_1}\chi_{P_i}(g_1)\cdot\chi_{P_{i'}}(g_1^{-1})\right)\cdot\left(\frac{1}{|G_2|}\sum_{g\in G_2}\chi_{Q_j}(g_2)\cdot\chi_{Q_{j'}}(g_2^{-1})\right)\\ &= (\chi_{P_i},\chi_{P_{i'}})\cdot(\chi_{Q_j},\chi_{Q_{j'}}).\end{aligned}$$

This shows that

$$(\chi_{P_i\otimes Q_j}, \chi_{P_{i'}\otimes Q_{j'}}) = \begin{cases} 1 & \text{if } P_i \cong P_{i'} \text{ and } Q_j \cong Q_{j'} \\ 0 & \text{else.} \end{cases}$$

In other words, each $P_i \otimes Q_j$ is irreducible and these representations are pairwise nonisomorphic. Since the number of conjugacy classes of $G_1 \times G_2$ is the product of the number of conjugacy classes of G_1 and G_2 respectively, we have found them all. Thus, the proof is completed. □

7. Lecture 7

7.1. **Application of number theory.** In this section k will denote an algebraically closed field of characteristic 0. The goal is to show that the dimension of an irreducible finite-dimensional representation divides the order of the group. First, we remind ourselves of some algebraic number theory.

Given a commutative ring R we say that $z \in R$ is *integral* over $\mathbb{Z}$ if there exists a nonconstant monic polynomial $f(x) \in \mathbb{Z}[x]$ such that $f(z) = 0$. In particular, if $R \supseteq \mathbb{C}$, we shall call z an *algebraic integer*.

Exercise 7.1. Show that $x \in \mathbb{Q}$ is an algebraic integer if and only if $x \in \mathbb{Z}$.

It is obvious that $\sqrt[3]{5}$ and $\sqrt{7}$ are algebraic integers but what about $\sqrt[3]{5} + \sqrt{7}$?

Proposition 7.2. *The following are equivalent for a commutative ring R.*

(i) $z \in R$ is integral over $\mathbb{Z}$,
(ii) The subring $\mathbb{Z}[z] \subseteq R$ is finitely generated as an abelian group,
(iii) $\mathbb{Z}[z]$ is contained in a finitely generated subgroup of R.

Proof. The fact that (ii) and (iii) are equivalent follows from the classification of finitely generated abelian group.

We will now show that (i) and (ii) are equivalent. If $z \in R$ satisfies a nonconstant monic polynomial $f(z) = 0$ of degree $n \geq 1$, then $\mathbb{Z}[z]$ is generated by 1, z, $\dots$, z^{n-1}, which can be proved using the division algorithm. This uses that f is monic.

Suppose that $\mathbb{Z}[z]$ is finitely generated. Then for some n we have that

$$\mathbb{Z}[z] = \langle 1, z, \dots, z^{n-1} \rangle.$$

Since $z^n \in \mathbb{Z}[z]$, we can find a nonconstant monic polynomial satisfied by z. □

Corollary 7.3. *The integral elements of R form a subring.*

Proof. Let $z_1, z_2 \in R$ be integral elements. Then, $\mathbb{Z}[z_1, z_2] \subseteq R$ is a finitely generated abelian group containing $\mathbb{Z}[z_1 \pm z_2]$ and $\mathbb{Z}[z_1 z_2]$. Proposition 7.2 implies that $z_1 \pm z_2$ and $z_1 z_2$ are integral. □

We remark that the above proof does not tell us which polynomials $z_1 + z_2$ and $z_1 z_2$ satisfy.

Proposition 7.4. *An element $\alpha \in \mathbb{C}$ is an algebraic integer if and only if it is the eigenvalue of a nonzero matrix with integer coefficients.*

Proof. If $\alpha \in \mathbb{C}$ is the eigenvalue of a nonzero matrix with integer coefficients, then it satisfies the characteristic polynomial.

If $\alpha \in \mathbb{C}$ is an algebraic integer it satisfies a polynomial equation

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0.$$

Consider the vector space $V = \mathbb{C}[x]/(f)$, on which multiplication by x gives rise to a linear operator $A : V \to V$. Furthermore,

$$f(x) = (x - z) \cdot g(x)$$

for some $g(x) \in \mathbb{C}[x]$ by the division algorithm. Hence, in $\mathbb{C}[x]/(f)$:

$$(A - z)(g(x)) = 0$$

and thus z is an eigenvalue. Explicitly, in the basis $\{1, x, \ldots, x^{n-1}\}$, the linear map A is given by the companion matrix

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}.$$

□

Thus, if z_1, z_2 are algebraic integers, we can find linear operators $A_1 : \mathbb{C}^{n_1} \to \mathbb{C}^{n_1}$ and $A_2 : \mathbb{C}^{n_2} \to \mathbb{C}^{n_2}$ with eigenvalues z_1 and z_2 respectively. Then

$$\begin{aligned} B &= A_1 \otimes \mathrm{id} + \mathrm{id} \otimes A_2 : \mathbb{C}^{n_1} \otimes \mathbb{C}^{n_2}, \\ C &= A_1 \otimes A_2, \end{aligned}$$

have eigenvalues $z_1 + z_2$ and $z_1 z_2$ respectively, which gives us an algorithmic way to find corresponding monic polynomials for sums and products.

We are now ready to prove the aforementioned theorem.

Theorem 7.5. *Suppose G is a finite group and that $\rho : G \to \mathrm{GL}(V)$ is a finite-dimensional irreducible representation of G over k. Then $n := \dim(V)$ divides $|G|$.*

Proof. By Schur's lemma,

$$z = \sum_g \chi_\rho(g^{-1})g$$

acts as a scalar $\lambda \cdot \mathrm{id}$ on V. Notice, that

$$\lambda = \frac{1}{n}\mathrm{Tr}(z) = \frac{1}{n}\sum_g \chi_\rho(g^{-1})\chi_\rho(g) = \frac{|G|}{n}.$$

This looks promising. □

Notice that each $\chi_\rho(g^{-1})$ is an algebraic integer since it is a sum of eigenvalues, each of which is a zero of $x^{|G|} - 1$. We will show that z is an algebraic integer and clearly it is sufficient to show that $\pi_C = \sum_{g \in C} g$ is an algebraic integer for each conjugacy class C. However, $\pi_C \in Z(\mathbb{Z}[G])$ which is a finitely generated abelian group. Thus, by Proposition 7.2, we deduce that z is an algebraic integer. However, we have a ring homomorphism taking $z \mapsto \lambda$ and the image of an algebraic integer in a ring homomorphism is again an algebraic integer. Since $\lambda \in \mathbb{Q}$ it therefore follows that $\lambda \in \mathbb{Z}$ and so $\dim(V)$ divides G, which was what we wanted to show.

Exercise 7.6. Show the following strengthened result. Let $Z \subseteq G$ be the center of G. Then $\dim(V)$ divides $[G : Z]$.

7.2. **Induced representations.** In this section, we will present yet another construction of representations. Suppose that $H \leq G$ are groups. Then, given a representation V of H, we can define a representation $\mathrm{Ind}_H^G(V) := k[G] \otimes_{k[H]} V$ of G which has the universal property that for any H-linear map $f : V \to W$ where W is a representation of G, there exists a unique G-linear map $\tilde{f} : \mathrm{Ind}_H^G(V) \to W$ making the following diagram commute:

$$\begin{tikzcd} V \arrow[r] \arrow[dr, "f"'] & \mathrm{Ind}_H^G(V) \arrow[d, dashed, "\tilde{f}"] \\ & W \end{tikzcd}$$

In a more general setting, given a ring homomorphism $A \to B$ we have two functors $\mathrm{Res}_H^G : \mathrm{Mod}(B) \to \mathrm{Mod}(A)$ and $\mathrm{Ind}_H^G : \mathrm{Mod}(A) \to \mathrm{Mod}(B)$ where $\mathrm{Ind}_H^G(M) = B \otimes_A M$ and these two functors are adjoint, i.e. for any A-module M and B-module N:

$$\mathrm{Hom}_B(\mathrm{Ind}_H^G(M), N) \cong \mathrm{Hom}_A(M, \mathrm{Res}_H^G(N)).$$

Exercise 7.7. We showed above that Ind_H^G is the left adjoint of Res_H^G. Show that the functor $\mathrm{Hom}_A(B, -)$ is the right adjoint of Res_H^G.

8. Lecture 8

8.1. **Frobenius theorem.** To understand representations V of a group G over a not necessarily algebraically closed field k, we want to understand $\mathrm{Hom}_G(V, V)$, which by Schur's lemma is a division algebra over k. Hence, we want to understand division algebras.

Theorem 8.1. *Any finite-dimensional division algebra over $\mathbb{R}$ is isomorphic to $\mathbb{R}$, $\mathbb{H}$ or $\mathbb{C}$.*

Proof. Let $\mathbb{R} \subseteq D$ be a finite-dimensional division algebra. Any element of D is the zero of an irreducible polynomial over $\mathbb{R}$, since D is finite-dimensional. The irreducible polynomials over $\mathbb{R}$ have degree 1 or 2. It is not hard to see that if $D \neq \mathbb{R}$, then there exists $i \in D$ such that $i^2 = -1$. Consider now the map

$$\begin{aligned} f : D &\longrightarrow D \\ i &\longmapsto ixi^{-1} \end{aligned}$$

which is a linear map. Furthermore, it clearly has eigenvalues in the set $\{\pm 1\}$. If all the eigenvalues of D are 1, then i is in the center of D and it follows that $D \cong \mathbb{C}$. If not, then we decompose $D = D_1 \oplus D_{-1}$ into nonempty eigenspaces. We can choose $j \in D_{-1}$ such that $j^2 = -1$, by an argument similar to before. Furthermore, $ij = -ji$. Lastly, D_1 is isomorphic to $\mathbb{C}$ and left multiplication by j gives a linear isomorphism between D_1 and D_{-1}. We conclude that D is 4-dimensional and since $i^2 = j^2 = -1$ and $ij = -ji$, we can conlude that $D \cong \mathbb{H}$. $\square$

9. Lecture 9

Recall that if $A \to B$ are rings, then we have a restriction functor $\mathrm{Res} : \mathrm{Mod}(B) \to \mathrm{Mod}(A)$ and a left-adjoint functor $\mathrm{Ind} : \mathrm{Mod}(A) \to \mathrm{Mod}(B)$ which takes M to $B \otimes M$. However, there is another natural choice of a functor, namely the right-adjoint to Res, which we shall call CoInd, defined by

$$\mathrm{CoInd}(M) = \mathrm{Hom}_A(B, M).$$

To check that this is a right adjoint, note that any map

$$f : N \to \mathrm{Hom}_A(B, M)$$

gives a map

$$\begin{aligned} f' &: N \to M \\ f'(v) &= f(v)(1) \end{aligned}$$

and conversely, any map

$$g : N \to M$$

gives a map

$$\begin{aligned} g' &: N \to \mathrm{Hom}_A(B, M) \\ g'(v)(b) &= g(bv). \end{aligned}$$

Exercise 9.1. Show that for $\mathbb{R} \subseteq \mathbb{C}$, Ind and CoInd are naturally isomorphic.

10. Lecture 10

In this lecture, we will specialise our result from the last lecture to the case of finite groups. Let $H \leq G$ be finite groups and thus $k[H] \subseteq k[G]$. Let V be a $k[H]$-module. Recall that

$$\begin{aligned} \mathrm{Ind}(V) &= k[G] \otimes_{k[H]} V, \\ \mathrm{CoInd}(V) &= \mathrm{Hom}_{k[H]}(k[G], V). \end{aligned}$$

Alternatively,

$$\mathrm{CoInd}(V) = \{f : G \to V \mid f(h \cdot g) = \rho_h(f(g)) \text{ for } g \in G, h \in H\}.$$

Example 10.1. As an example, let $V = k$. Then,

$$\mathrm{CoInd}(V) = \{f : H \backslash G \to k\} = k[H \backslash G].$$

We cna describe $\mathrm{Ind}(V)$ explicitly, by choosing $g_i \in G$ for which

$$G = \cup_i g_i g_i H.$$

Then,

$$k[G] = \oplus_{i \in I} g_i k[H]$$

and thus as a vector space

$$k[G] \otimes_{k[H]} V = \oplus_i g_i \otimes V.$$

$y = x$ $\qquad$ $y = x^2$

FIGURE 1. Examples of algebraic varieties

For finite groups $H \leq G$, giving a G-homomorphism $\mathrm{Ind}(V) \to \mathrm{CoInd}(V)$ is equivalent to giving a H-homomorphism from V to $\mathrm{CoInd}(V)$ by the adjoint property. Hence, to see that CoInd and Ind are naturally isomorphic, we note that we have a map:

$$\begin{aligned} V &\longrightarrow \mathrm{CoInd}(V) \\ v &\longmapsto \left(f_v : G \to V, \quad f_v(g) = \begin{cases} \rho_g(v), & g \in H \\ 0, & g \notin H. \end{cases} \right) \end{aligned}$$

11. Lecture 11

11.1. **Motivation.** So far, we have only studied the representation theory of finite groups. However, we would like to be able to study representations of infinite groups such as $\mathbb{C}^\times$ and $\mathrm{GL}_n(\mathbb{C})$ as well.

Example 11.1. What are the 1-dimensional representations of $\mathbb{C}^\times$? These are simply the group homomorphisms $\rho : \mathbb{C}^\times \to \mathbb{C}^\times$. However, using the axiom of choice and the algebraic closure of $\mathbb{C}$, one can show that there are infinitely many field automorphisms of $\mathbb{C}$, each of which induces a group homomorphism $\mathbb{C}^\times \to \mathbb{C}^\times$. At this point a physicist would complain that most of these representations are unphysical. A mathematician, on the other hand, can object that the class of 1-dimensional representations is too large to say something meaningful in general. Hence, both the mathematician and the physicist have to impose additional constraints on the representations to get something meaningful. For instance, if we restrict our attention to 1-dimensional representations $\rho : \mathbb{C}^\times \to \mathbb{C}^\times$ that are rational functions, i.e. of the form $\rho(z) = \frac{f(z)}{z^n}$, then since $\mathbb{C}$ is algebraically closed, $f(z) = az^n \in \mathbb{C}^\times$. For ρ to be multiplicative, we need $a = 1$. Thus, the algebraic 1-dimensional representations of $\mathbb{C}$ are precisely the maps $z \mapsto z^n$ for some $n \in \mathbb{N}$.

In this lecture, we will try to formalise what an algebraic representation should mean in general.

11.2. **Algebraic geometry.** In this section, we will introduce the necessary background from algebraic geometry. The main idea is that we want to study solution sets of polynomials, which informally will be *algebraic varieties*. Two examples are given in Figure 1. Geometrically, we want to say that the examples in Figure 1 are "the same". Hence, we must ask ourselves what geometric data we want to remember. A reasonable answer is given by the following definition.

Definition 11.2. Fix a field k. An *(affine) algebraic variety* over k is a commutative k-algebra A.

Given a k-algebra A, we denote its corresponding variety by $\mathrm{Spec}(A)$.

Example 11.3. Given a collection of polynomials $f_1, \dots, f_n \in k[x_1, \dots, x_m]$ we can consider the algebraic variety

$$\mathrm{Spec}(k[x_1, \dots, x_m]/(f_1, \dots, f_n)).$$

Since

$$\mathbb{C}[x, y]/(y - x) \cong \mathbb{C}[x] \cong \mathbb{C}[x, y]/(y - x^2)$$

we can justify that the geometry in Figure 1 is "the same".

Given the zero set of some polynomial equations, we could associate an affine algebraic variety. How can we reconstruct some "zero set" from an arbitrary commutative k-algebra?

Example 11.4. As an example, consider $A = \mathbb{C}[x, y]/(f)$ where $f(x) \in k[x]$. Notice that by the universal property of a quotient, $\mathrm{Hom}_{\mathbb{C}}(A, \mathbb{C})$ is in bijection with the zero set of f in $\mathbb{C}^2$. This motivatives the following definition.

Definition 11.5. The k-points of an affine algebraic variety $X = \mathrm{Spec}(A)$ is defined to be

$$X(k) = \mathrm{Hom}_k(A, k).$$

We must now define the morphisms in the category of affine algebraic varieties over k. In particular, we want a morphism of affine algebraic varieties to give a morphism of the k-points. Hence, we may want to define the category of affine algebraic varieties as the *opposite category* of the category of commutative k-algebras. In particular, a morphism $X = \mathrm{Spec}(A) \to \mathrm{Spec}(B) = Y$ gives rise to a map of points $X(k) \to Y(k)$.

Since the coproduct in the category CAlg_k is the tensor product, it follows that if $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$, then $X \times Y = \mathrm{Spec}(A \otimes_k B)$. In particular, this gives us that

$$(X \times Y)(k) = \mathrm{Hom}_k(\mathrm{Spec} k, X \times Y) = \mathrm{Hom}_k(\mathrm{Spec} k, X) \times \mathrm{Hom}_k(\mathrm{Spec} k, Y) = X(k) \times Y(k).$$

Example 11.6. Let $R = k[x_1, \dots, x_n]$. Then, we let $\mathbb{A}^n := \mathrm{Spec}(R)$. Notice that

$$\mathbb{A}^n(k) = k^n.$$

11.3. **Algebraic groups.** With the preliminaries from algebraic geometry out of the way, we can now define what an algebraic group is.

Definition 11.7. An *(affine) algebraic group* is an affine algebraic variety G with three maps

$$\begin{aligned} m : G \times G &\longrightarrow G, \\ \iota : G &\longrightarrow G, \\ e : \mathrm{Spec}(k) &\longrightarrow G, \end{aligned}$$

such that the following diagrams commute:

(i) **Associativity:**

$$\begin{tikzcd} G \times G \times G \arrow[r, "\mathrm{id} \times m"] \arrow[d, "m \times \mathrm{id}"'] & G \times G \arrow[d, "m"] \\ G \times G \arrow[r, "m"'] & G \end{tikzcd}$$

(ii) **Identity:**

$$\begin{tikzcd} \mathrm{Spec}(k) \times G \arrow[r, "\simeq"] \arrow[d, "e\times\mathrm{id}"'] & G \arrow[d, "\mathrm{id}"] \\ G \times G \arrow[r, "m"] & G \end{tikzcd} \qquad \begin{tikzcd} G \times \mathrm{Spec}(k) \arrow[r, "\simeq"] \arrow[d, "\mathrm{id}\times e"'] & G \arrow[d, "\mathrm{id}"] \\ G \times G \arrow[r, "m"] & G \end{tikzcd}$$

(iii) **Inverses:**

$$\begin{tikzcd} G \arrow[r, "(\mathrm{id},\iota)"] \arrow[d] & G \times G \arrow[d, "m"] \\ \mathrm{Spec}(k) \arrow[r, "e"'] & G \end{tikzcd} \qquad \begin{tikzcd} G \arrow[r, "(\iota,\mathrm{id})"] \arrow[d] & G \times G \arrow[d, "m"] \\ \mathrm{Spec}(k) \arrow[r, "e"'] & G \end{tikzcd}$$

Exercise 11.8. Show that if G is an algebraic group, then $G(k)$ is a group in the usual sense.

Example 11.9. Consider the algebraic group $\mathbb{G}_a = \mathrm{Spec}(k[x])$ with multiplication map induced by

$$\begin{aligned} k[x] &\longrightarrow k[x,y] \\ x &\longmapsto x+y. \end{aligned}$$

Then, the induced group structure on $\mathbb{G}_a(k) = k$ is given by field addition in k.

Example 11.10. Consider the algebraic group $\mathbb{G}_m = \mathrm{Spec}(k[x,x^{-1}])$ with multiplication map induced by

$$\begin{aligned} k[x,x^{-1}] &\longrightarrow k[x,x^{-1}] \otimes k[x,x^{-1}] \\ x &\longmapsto x \otimes x. \end{aligned}$$

Then, we see that for any k-algebra A,

$$\mathrm{Mor}(\mathrm{Spec}(A), \mathbb{G}_m) \cong A^\times$$

and the induced group operation is multiplication.

Example 11.11. For $\mathrm{GL}(n) := \mathrm{Spec}\left(k\left[a_{ij}, \frac{1}{\det(a_{ij})}\right]\right)$ we have that $\mathrm{GL}(n)(k) \cong \mathrm{GL}_n(k)$ and if $\mathcal{O}(\mathrm{GL}(n)) = k\left[a_{ij}, \frac{1}{\det(a_{ij})}\right]$, then we have a multiplication map induced by

$$\begin{aligned} \mathcal{O}(\mathrm{GL}(n)) &\longrightarrow \mathcal{O}(\mathrm{GL}(n)) \otimes \mathcal{O}(\mathrm{GL}(n)) \\ a_{ij} &\longrightarrow f_{ij}(a_{i'j'}, b_{i'',j''}) \end{aligned}$$

where f_{ij} are polynomials such that $(a_{ij})(b_{ij}) = (f_{ij}(a_{i'j'}, b_{i''j''}))$.

Definition 11.12. A representation of an algebraic group G is a morphism of algebraic varieties

$$\rho : G \to \mathrm{GL}(n)$$

that is compatible with the commutative diagrams in the definition of an algebraic group.

As should be evident by now, algebraic groups are complicated objects that package a lot of data. To study them, we will need to understand Lie algebras.

11.4. **Tangent space to an algebraic variety.** Recall that if we have a smooth submersion $f : \mathbb{R}^2 \to \mathbb{R}$, then the tangent space to the zero set of f at (x_0, y_0) is given by all points $(x, y) \in \mathbb{R}^2$ such that

$$f_x(x_0, y_0) \cdot (x - x_0) + f_y(x_0, y_0) \cdot (y - y_0) = 0.$$

Recall also that

$$f(x, y) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0) + \text{higher order terms}.$$

From this, it is reasonable to give the following definition.

Definition 11.13. Let X be an algebraic variety and $a \in X(k)$. Then, the *tangent space* to X at a is defined as

$$T_{X,a} \coloneqq \{f : \mathrm{Spec}(k[\epsilon]/(\epsilon^2)) \to X : f|_{\mathrm{Spec}(k)} = a\}.$$

12. Lecture 12

Recall that an algebraic group is an affine scheme G together with three maps

$$\begin{aligned} m : G \times G &\longrightarrow G, \\ \iota : G &\longrightarrow G, \\ e : \mathrm{Spec}(K) &\longrightarrow G, \end{aligned}$$

satisfying certain properties.

Given an affine scheme $X = \mathrm{Spec}(A)$, then for any k-algebra B, we can consider the B-points:

$$X(B) = \mathrm{Mor}(\mathrm{Spec}(B), X) = \mathrm{Hom}_k(A, B).$$

Example 12.1. If $k = \mathbb{R}$ and $X = \mathrm{Spec}(\mathbb{R}[x, y]/(x^2 + y^2 + 1))$, then

$$X(\mathbb{R}) = \mathrm{Hom}_{\mathbb{R}}\left(\mathbb{R}[x, y]/(x^2 + y^2 + 1)\right) = \emptyset$$

while $X(\mathbb{C})$ is large. In conclusion, there are interesting schemes with no points.

By the Yoneda embedding, any affine scheme $X = \mathrm{Spec}(A)$ is a functor $X : \mathrm{CAlg}_k \to \mathrm{Set}$, given by

$$X(B) = \mathrm{Hom}_k(A, B).$$

In particular, the following exercise follows.

Exercise 12.2. If G is an affine group scheme, then $G(B)$ is either empty or a group.

Given an affine group scheme G, we define its ring of functions as the unique k-algebra $\mathcal{O}(G)$ such that $G = \mathrm{Spec}(\mathcal{O}(G))$. Given any $f \in \mathcal{O}(G)$, we obtain a map

$$\begin{aligned} G(k) &\longrightarrow k, \\ (g : \mathcal{O}(G) \to k) &\longmapsto g(f). \end{aligned}$$

Since we have an equivalence of categories between affine schemes over k and k-algebras, it follows that we have maps

$$\begin{aligned} \Delta : \mathcal{O}(G) &\longrightarrow \mathcal{O}(G) \times \mathcal{O}(G) && \text{comultiplication} \\ \epsilon : \mathcal{O}(G) &\longrightarrow k && \text{counit} \\ \iota^* : \mathcal{O}(G) &\longrightarrow \mathcal{O}(G) && \text{antipode.} \end{aligned}$$

From the commutative diagrams of Δ and ϵ we get a structure called a coalgebra.

Example 12.3. By dualising, we obtain maps

$$\begin{aligned}\Delta^* : \mathcal{O}(G)^* \otimes \mathcal{O}(G)^* &\longrightarrow \mathcal{O}(G)^* \\ \epsilon^* : k &\longrightarrow \mathcal{O}(G)^*.\end{aligned}$$

However, to obtain the map Δ^*, we must embed $V_1^* \otimes V_2^* \hookrightarrow (V_1 \otimes V_2)^*$, which we can do, but for infinite dimensional vector spaces, this might not be a linear isomorphism. This means that we lose some data in this construction.

One can verify that $(\mathcal{O}(G)^*, \Delta^*, \epsilon^*)$ is an algebra. Our previous loss of data shows that the coalgebra structure is an algebra structure with some additional data.

Example 12.4. Let Γ be a finite group and define

$$\mathcal{O}(\underline{\Gamma}) \coloneqq \text{algebra of functions } \Gamma \to k.$$

Consider the algebraic variety $\underline{\Gamma} = \operatorname{Spec}(\mathcal{O}(\Gamma))$. Then, we have a basis of $\mathcal{O}(\underline{\Gamma})$ given by δ_g for each $g \in \Gamma$, with multiplication

$$\delta_g \cdot \delta_g' = \begin{cases} \delta_g & g = g', \\ 0 & g \neq g. \end{cases}$$

This implies that we have maps

$$\begin{aligned}\Delta : \mathcal{O}(\underline{\Gamma}) &\longrightarrow \mathcal{O}(\underline{\Gamma}) \otimes \mathcal{O}(\underline{\Gamma}) \\ \delta_g &\longmapsto \sum_{g_1 g_2 = g} \delta_{g_1} \otimes \delta_{g_2} \\ \epsilon : \mathcal{O}(\underline{\Gamma}) &\longrightarrow k \\ f &\longmapsto f(1)\end{aligned}$$

that impose a coalgebra structure on $\mathcal{O}(\underline{\Gamma})$.

Exercise 12.5. Prove that $\mathcal{O}(\underline{\Gamma})^* \cong k[\Gamma]$.

Example 12.6. In this example, we show that not all finite group schemes arise as in the previous example. Let $n > 1$ and consider

$$\mu_n(B) \coloneqq \{\text{solutions to } z^n = 1 \text{ in } B^\times\} \subseteq B^\times.$$

Explicitly

$$\mathcal{O}(\mu_n) = k[x, x^{-1}]/(x^n - 1).$$

We define $\Delta(x^m) = x^m \otimes x^m$.

If $n = 2$ and $k = \mathbb{F}_2$, then

$$\dim(\mathcal{O}(\mu_2)^*) = \dim_k(k[x]/(x^2 - 1)) = \boxed{2}$$

while

$$|\mu_n(\mathbb{F}_2)| = \boxed{1}.$$

If this example came from a finite group Γ, these two numbers would have been equal.

13. Lecture 13

13.1. **Yoneda's lemma.**

Exercise 13.1. What is the center of the noncommutative ring of differential operators $D_{\mathbb{A}^1_k} = k\langle x, \partial_x\rangle/(\partial_x x - x\partial_x - 1)$?

Recall that given an affine scheme X, we obtain a functor

$$\begin{aligned} \mathrm{CAlg} &\longrightarrow \mathrm{Set} \\ B &\longmapsto X(B). \end{aligned}$$

The special case of a result from category theory tells us that the scheme is uniquely defined by this functor.

Lemma 13.2 (Special case of Yoneda)**.** *Let* $X = \mathrm{Spec}(A)$ *and* $X' = \mathrm{Spec}(A')$ *be affine schemes. If the induced functors of points are naturally isomorphic, then* $X \cong X'$*.*

Proof. Suppose that there exists a natural transformation with bijective components $\phi_B : X(B) \to X'(B)$. Define

$$\begin{aligned} f &\coloneqq \phi_{A'}^{-1} : A \to A', \\ g &\coloneqq \phi_A : A' \to A. \end{aligned}$$

Since ϕ is a natural transformation, the map f gives as a commutative diagram:

$$\begin{tikzcd} X(A) \arrow[r, "\phi_A"] \arrow[d, "X(f)"'] & X'(A) \arrow[d, "X'(f)"] \\ X(A') \arrow[r, "\phi_{A'}"'] & X'(A') \end{tikzcd}$$

Since

$$(X'(f) \circ \phi_A)(\mathrm{id}) = X'(f)(g) = f \circ g$$

and

$$(\phi_{A'} \circ X(f))(\mathrm{id}) = \phi_{A'}(f) = \mathrm{id}$$

it follows from the commutativity that $f \circ g = \mathrm{id}$. Similarly, $g \circ f = \mathrm{id}$. It follows that A and A' are isomorphic rings.

Yoneda's lemma says that affine schemes embeds into the category $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Set})$, but for algebraic groups this can be upgraded to an embedding into $\mathrm{Fun}(\mathrm{CAlg}_k, \mathrm{Grp})$. □

13.2. **More on representations of algebraic groups.** We have defined what a representation of an algebraic group is, but this is a hard definition to work with in practise. However, given a representation $\rho : G \to \mathrm{GL}(n)$ of an affine group scheme G, we obtain from the functors of points view a map

$$G(\mathcal{O}(G)) \longrightarrow \mathrm{GL}_n(\mathcal{O}(G))$$

that takes id to a map

$$\rho_{\mathcal{O}(G)}(\mathrm{id}) : k^n \longrightarrow k^n \otimes \mathcal{O}(G).$$

Proposition 13.3. *The data of a representation* $\rho : G \to \mathrm{GL}(n)$ *is equivalent to the data of a k-linear maps*

$$a : k^n \longrightarrow k^n \otimes \mathcal{O}(G)$$

such that the following diagrams commute:

(i) $k^n \xrightarrow{a} k^n \otimes \mathcal{O}(G) \overset{a\otimes\mathrm{id}}{\underset{\mathrm{id}\otimes\Delta}{\rightrightarrows}} k^n \otimes \mathcal{O}(G) \otimes \mathcal{O}(G)$

(ii)
$$\begin{tikzcd} k^n \arrow[r, "a"] \arrow[d, "\mathrm{id}"'] & k^n \otimes \mathcal{O}(G) \arrow[d, "\epsilon"] \\ k^n \arrow[r, "\simeq"'] & k^n \otimes k \end{tikzcd}$$

Corollary 13.4. *If Γ is a finite group and*

$$G := \mathrm{Spec}(k\textit{-valued functions on } \Gamma)$$

then since $\dim(\mathcal{O}(G)) < \infty$ it follows that the data of a representation is equivalent to a $\mathcal{O}(G)^*$-module structure on k^n. We know that $\mathcal{O}(G)^* \cong k[\Gamma]$ and thus the theory of representations of finite groups is included in the theory of algebraic groups.

We will now give the definition of a representation of G on any vector space.

Definition 13.5. A representation of an affine group scheme G on a vector space V over k is a comodule structure on V, i.e. a map

$$a : V \longrightarrow V \otimes \mathcal{O}(G)$$

such that

(i) $(\mathrm{id} \times \delta) \circ a = (a \circ \mathrm{id}) \circ a$,
(ii) $(\mathrm{id} \otimes \epsilon) \circ a = \mathrm{id}$.

Note that if V is finite-dimensional, the above is the same as a $\mathcal{O}(G)^*$-module structure on V.

Theorem 13.6. *A representation of $\mathbb{G}_m := \mathrm{Spec}(k[t, t^{-1}])$ on V is a $\mathbb{Z}$-grading on V. The irreducible representations are exactly given by the map $\mathbb{G}_m \to \mathbb{G}_m$ defined pointwise by*

$$\begin{aligned} \mathbb{G}_m(B) &\longrightarrow \mathbb{G}_m(B) \\ z &\longmapsto z^n \end{aligned}$$

for every $n \in \mathbb{Z}$.

Proof. We will only sketch a proof. Given a map

$$a : V \to V \otimes k[t, t^{-1}] = \bigoplus_{i \in \mathbb{Z}} V \otimes t^i$$

we obtain projection maps

$$\pi_n : V \to V \otimes t^n \cong V.$$

It follows from the axioms of a comodule structure that $\pi_n^2 = \pi_n$, $\pi_n \circ \pi_m = 0$ for $n \neq m$ and $\sum_n \pi_n = \mathrm{id}$. From this, we obtain a $\mathbb{Z}$-grading on V characterising the representation up to isomorphism. □

13.3. **Tangent space.** Recall the definition of a tangent space from an earlier class. It is not immediate why the tangent space has a natural structure of a vector space. To prove that it does suppose that $\varphi : \mathrm{Spec}\left(k[\epsilon]/(\epsilon^2)\right) \to X$ is in T_{x,x_0}. Then, by definition

$$\varphi^*(a) = \mathrm{ev}_{x_0}(a) + D(a) \cdot \epsilon$$

for every $a \in A = \mathcal{O}(G)$, for some $D(a) \in k$. It is not hard to see that the map $D : A \to k$ makes φ^* into a map of k-algebras if and only if it is linear and satisifies *Leibniz formula*:

$$D(ab) = D(a) \cdot \mathrm{ev}_{x_0}(b) + \mathrm{ev}_{x_0}(a) \cdot D(b)$$

for all $a, b \in A$.

Example 13.7. Let $X = \mathbb{A}^n$ with a point $0 \in \mathbb{A}^n(k)$. Then any derivation

$$D : k[x_1, \ldots, x_n] \to k$$

at 0 is a directional derivative.

14. Lecture 14

Let us begin by studying some examples of tangent spaces, as defined in the previous lecture. We will denote the *dual numbers* by

$$\Lambda := k[\epsilon]/(\epsilon^2).$$

Example 14.1. Consider an affine scheme X of the form $\mathcal{O}(X) = k[x_1, \ldots, x_n]/(h)$. This is a so-called *hypersurface* in $\mathbb{A}^n$ and

$$X(k) = \{x \in \mathbb{A}^n : h(x) = 0\}.$$

Fix any $a \in X(k)$. Then, by definition, finding the tangent space $T_{X,a}$ is equivalent to solving for $a_i + b_i\epsilon \in \Lambda$ such that $h(a + b\epsilon) = 0$. We know from calculus that

$$h(x) = h(a) + \sum_i \frac{\partial h}{\partial x_i}(a) \cdot (x_i - a_i) + (x_1 - a_1, \ldots, x_n - a_n)^2.$$

Thus, it is a straightforward computation to see that

$$h(a + b\epsilon) = \epsilon \sum_i \frac{\partial h}{\partial x_i}(a) \cdot b_i.$$

In conclusion,

$$T_{X,a} = \left\{ b \in k^n : \sum_i \frac{\partial h}{\partial x_i}(a) \cdot b_i = 0 \right\}$$

as expected from multivariable calculus.

In the following examples, we will get a vector space that will eventually be equipped with the structure of a *Lie algebra*.

Example 14.2. Let $X = \mathrm{GL}(n)$. Then

$$T_{X,\mathrm{id}} = \{\mathrm{id} + \epsilon B \in \mathrm{GL}_n(\Lambda) : B \in M_n(k)\} = M_n(k)$$

since for any $B \in M_n(k)$,

$$(\mathrm{id} + \epsilon B)(\mathrm{id} - \epsilon B) = \mathrm{id}$$

in $M_n(\Lambda)$.

Example 14.3. Let $X = O(n)$. Then

$$T_{X,\mathrm{id}} = \{\mathrm{id} + B\epsilon \in O_n(\Lambda) : B \in M_n(k)\} = \text{skew-symmetric matrices over } k,$$

since

$$(\mathrm{id} + B\epsilon)^t(\mathrm{id} + B\epsilon) = \mathrm{id} \iff \epsilon(B + B^t) = 0 \iff B^t = -B.$$

Example 14.4. Consider $\mu_n \subseteq \mathbb{G}_m$, given by $\mu_n = \mathrm{Spec}(k[x]/(x^n - 1))$. Then

$$T_{\mu_n,1} = \{(1+\epsilon z)^n = 1 : z \in k\} = \begin{cases} k & \text{if } \mathrm{char}(k) \mid n, \\ 0 & \text{else,} \end{cases}$$

since

$$1 = (1+\epsilon z)^n \iff nz\epsilon = 0 \iff nz = 0.$$

This is saying that μ_n is discrete whenever $\mathrm{char}(k) \nmid n$.

Given an algebraic group G, we define

$$\mathrm{Lie}(G) \coloneqq T_{G,e}.$$

We now want to explore what a vector field should mean in the algebraic geometry setting.

Definition 14.5. Let A be any k-algebra. Then, we define a *derivation* of A to be a k-linear map $d : A \to A$ such that

$$d(ab) = d(a)b + ad(b)$$

for all $a, b \in A$.

Note that derivations form a vector space.

Exercise 14.6. Show that if $d_1, d_2 : A \to A$ are derivations of A, then $[d_1, d_2] \coloneqq d_1 d_2 - d_2 d_1$ is also a derivation of A.

Definition 14.7. A *vector field* on an affine scheme X is by definition a derivation of $\mathcal{O}(X)$.

Note that the derivations of $\mathcal{O}(X)$ form a $\mathcal{O}(X)$-module.

Intuitively, we want a vector field to give us a tangent vector at each point. To see that the above definition is reasonable, we observe that given a derivation $d \in \mathrm{Der}(\mathcal{O}(X))$ and a point $a \in X(k)$, we have $ev_a \circ d \in T_{X,a}$.

Definition 14.8. The *Lie derivative* of a function $f \in \mathcal{O}(X)$ along a vector field $d : \mathcal{O}(X) \to \mathcal{O}(X)$ is defined to be $d(f) \in \mathcal{O}(X)$.

Exercise 14.9. Let $A = k[x]$. Prove that $\mathrm{Der}(A) = \{f \frac{\partial}{\partial x}\}$ and that

$$\left[f\frac{\partial}{\partial x}, g\frac{\partial}{\partial x}\right] = (fg' - gf')\frac{\partial}{\partial x}.$$

15. Lecture 15

The reference for much of this treatment is *Abelian varieties* by Mumford.

Example 15.1. Let $G = \mathrm{SL}(n)$. Then

$$\begin{aligned} \mathrm{Lie}(G) &= \ker(G(\Lambda) \to G(k)) \\ &= \{1 + \epsilon A : \det(1+\epsilon A) = 1, A \in M_n(k).\} \end{aligned}$$

If $A = (a_{ij})$, then

$$\det(1+\epsilon A) = 1 \iff \prod_i (1 + \epsilon a_{ii}) = 1 \iff \mathrm{Tr}(A) = 0.$$

Thus,

$$\mathfrak{sl}(n) \coloneqq \mathrm{Lie}(G) = \{A \in M_n(k) : \mathrm{Tr}(A) = 0\}.$$

Definition 15.2. A *Lie algebra* $\mathfrak{g}$ over a field k is a vector space $\mathfrak{g}$ over k with a map

$$[-,-] : \Lambda^2\mathfrak{g} \longrightarrow \mathfrak{g}$$

satisfying the *Jacobi identity*:

$$[a,[b,c]] = [[a,b],c] + [b,[a,c]].$$

Example 15.3. Let A be an associative algebra over k and define $[a,b] = ab - ba$. Then A together with $[-,-]$ is a *Lie algebra.* For all $a \in A$ we can define the map

$$\mathrm{ad}_a(b) = [a,b].$$

One can check that ad_a is a derivation, which is equivalent to the fact that $[-,-]$ satisfies the Jacobi identity.

Example 15.4. Some examples of Lie algebras are

- $\mathfrak{gl}(n) = M_n(k)$
- $\mathfrak{sl}(n) =$ traceless elements of $M_n(k)$
- $\mathfrak{o}(n) =$ skew-symmetric elements of $M_n(k)$

Example 15.5. Let A be an associative algebra. Then $\mathrm{End}_k(A)$ together with the Lie bracket

$$[A,B] = A \circ B - B \circ A$$

and $\mathrm{Der}(A)$ is a subalgebra. Our goal is to make $\mathrm{Lie}(G)$ into a Lie algebra by embedding it in $\mathrm{Der}(\mathcal{O}(G))$.

Definition 15.6. A derivation $D \in \mathrm{Der}(\mathcal{O}(G))$ is called *left-invariant* if the diagram

$$\begin{tikzcd}
\mathcal{O}(G)\otimes\mathcal{O}(G) \arrow[r, "\mathrm{id}\otimes D"] & \mathcal{O}(G)\otimes\mathcal{O}(G) \\
\mathcal{O}(G) \arrow[u, "\Delta"] \arrow[r, "D"'] & \mathcal{O}(G) \arrow[u, "\Delta"']
\end{tikzcd}$$

commutes.

Our intuition from Lie groups as they appear in differential topology is that left-invariant vector fields are determined by tangent vectors in $T_{G,e}$. The following theorem says that this still holds in the algebraic setting.

Exercise 15.7. Left-invariant vector fields are closed under composition.

Theorem 15.8. *The map*

$$\{\textit{Left-invariant vector fields on } G\} \longrightarrow \mathrm{Lie}(G)$$

$$\varphi : D \longmapsto (d : \mathcal{O}(G) \xrightarrow{D} \mathcal{O}(G) \xrightarrow{\mathrm{ev}_e} k)$$

is an isomorphism of vector spaces. Since left-invariant vector fields are closed under composition this allows us to give the structure of a Lie algebra to $\mathrm{Lie}(G)$.

Let us begin with some examples.

Example 15.9. Notice that

$$\mathrm{Der}(\mathcal{O}(\mathbb{G}_a)) = \mathrm{Der}(k[x]) = \{f \cdot \frac{\partial}{\partial x}\}.$$

A derivation $f(x) \cdot \frac{\partial}{\partial x}$ is left-invariant if and only if for all $h(x) \in k[x]$:

$$f(y) \cdot h'(x+y) = f(x+y) \cdot h'(x+y)$$

holds in $k[x,y]$. This translates to the fact that $f(x+y) = f(y)$ which gives that f is a constant. Thus, the Lie algebra of $\mathbb{G}_a$ is k as a vector space with trivial Lie bracket.

Example 15.10. We have that

$$\mathrm{Der}(\mathcal{O}(\mathbb{G}_m)) = \{f\frac{\partial}{\partial x}\}.$$

The derivation $f(x) \cdot \frac{\partial}{\partial x}$ is left-invariant if and only if for all $h(x) \in k[x]$:

$$xf(y)h'(xy) = f(xy)h'(xy)$$

holds in $k[x,x^{-1},y,y^{-1}]$, which is true if and only if $f(xy) = xf(y)$, i.e. $f(x) = cx$ for some constant $c \in k$. Again, the Lie algebra is k with trivial Lie bracket.

Notice that

$$\mathrm{Aut}_\Lambda(\mathcal{O}_G \otimes \Lambda) \supseteq \{\varphi : \mathcal{O}(G) \otimes \Lambda \to \mathcal{O}(G) \otimes \Lambda : \varphi = \mathrm{id} \mod \epsilon\} \cong \mathrm{Der}(\mathcal{O}(G)).$$

We can now prove the previous theorem.

Proof. We will define explicitly an inverse map. Suppose we have $d : \mathrm{Spec}(\Lambda) \to G$ such that

$$\begin{tikzcd} \mathrm{Spec}(\Lambda) \arrow[r, "d"] & G \\ \mathrm{Spec}(k) \arrow[u] \arrow[ur, "e"'] & \end{tikzcd}$$

commutes. Define the map

$$\varphi : G \times \mathrm{Spec}(\Lambda) \xrightarrow{(\mathrm{id}\times d, \pi_2)} G \times G \times \mathrm{Spec}(\Lambda) \xrightarrow{m\times\mathrm{id}} G \times \mathrm{Spec}(\Lambda).$$

We leave it as an exercise to verify that we have produced mutual inverses. $\square$

Example 15.11. Let $G = \mathrm{GL}(n)$ and thus $\mathrm{Lie}(G) = M_n(k)$. If $A \in M_n(k)$, then the corresponding map φ is given by $\varphi(B) = B(1 + \epsilon A)$.

Let us finish with another interpretation of the Lie bracket on $\mathrm{Lie}(G)$. Suppose that we have two elements $d_i \in \mathrm{Lie}(G)$ for $i = 1, 2$. Then, we obtain a map

$$\mathrm{Spec}(\Lambda_1) \times \mathrm{Spec}(\Lambda_2) \xrightarrow{d\times d'} G \times G \xrightarrow{\text{group commutator}} G$$

which gives a map $\mathcal{O}(G) \to k[\epsilon_1, \epsilon_2]/(\epsilon_1^2, \epsilon_2^2)$. Projecting down to $k[\epsilon\epsilon']/(\epsilon\epsilon')$ gives us $[d, d']$.

Example 15.12. Consider $G = \mathrm{GL}(n)$ and $\mathrm{Lie}(G) = \{1 + A\epsilon\}$. Notice that

$$\begin{aligned}(1+\epsilon A)(1+\epsilon' A')(1-\epsilon A)(1-\epsilon' A') &= (1+\epsilon A+\epsilon' A'+\epsilon\epsilon' AA')(1-\epsilon A-\epsilon' A'+\epsilon\epsilon' AA') \\ &= 1+\epsilon\epsilon'[A,A'].\end{aligned}$$

We remark that in general, if G is abelian, then $[-,-]$ is trivial.

16. Lecture 16

We will start with some recollection. If G is an algebraic group with identity $e \in G(k)$, then
$$\mathrm{Lie}(G) = T_{G,e}$$
is a Lie algebra. We have two ways to define the Lie bracket.

- Recall that
$$T_{G,e} \cong \{\text{left-invariant derivations } D : \mathcal{O}(G) \to \mathcal{O}(G)\}.$$
Since left-invariant derivations are closed under the Lie bracket, we obtain a Lie bracket on $T_{G,e}$.
- The other definition, is more local and easier to compute with. Let $\Lambda_i = k[\epsilon_i]/(\epsilon_i^2)$ for $i = 1, 2$ and take $d_1, d_2 \in T_{G,e}$. We have projection maps $p_i : \mathrm{Spec}(\Lambda_1) \times \mathrm{Spec}(\Lambda_2) \to \mathrm{Spec}(\Lambda_i)$. Then, we can define
$$[d_1, d_2] = (d_1 \circ p_1)(d_2 \circ p_2)(d_1 \circ p_1)^{-1}(d_2 \circ p_2)^{-1} \in G(k[\epsilon_1\epsilon_2]/(\epsilon_1^2\epsilon_2^2)) \cong G(\Lambda).$$

Example 16.1. Let $G = \mathrm{GL}(n)$. Then, if $d_1 = 1 + \epsilon A, d_2 = 1 + \epsilon' B \in \mathrm{Lie}(G)$, we have that
$$(1 + \epsilon A)(1 + \epsilon' B)(1 - \epsilon A)(1 - \epsilon' B) = 1 + [A, B]\epsilon\epsilon',$$
which proves that the Lie algebra $\mathfrak{gl}(n)$ is $M_n(k)$ with the standard Lie bracket.

17. Digression: non-affine group schemes

In this course, we are only considering affine group schemes, but let us take a moment to think about an example of a non-affine group scheme. Let k be a field such that $\mathrm{char}(k) \neq 2$ and consider elliptic curves
$$y^2 = x^3 + ax + b$$
with $a, b \in \mathbb{Z}$ and $x^3 + ax + b$ with no multiple roots.

Exercise 17.1. Show that the points on elliptic curves over $\mathbb{R}$ are smooth manifolds.

We can homogenise the elliptic curve as
$$y^2 z = x^3 + axz^2 + bz^3.$$
Then, we define
$$E(R) \coloneqq \{[x_0 : y_0 : z_0] \in \mathbb{P}^2(R) : y_0^2 z_0 = x_0^3 + ax_0z_0^2 + bz_0^3\}$$
where $\mathbb{P}^2(R)$ is the set of equivalence classes of tuples $(x_0, y_0, z_0) \in R^3$ such that $(1) = (x_0, y_0, z_0)$ and where the equivalence is defined by
$$(x_0, y_0, z_0) \sim (\lambda x_0, \lambda y_0, \lambda z_0)$$
for $\lambda \in R^\times$. We see that
$$E(k) = \{\text{solutions to } y^2 = x^3 + ax + b\} \cup \{\infty\}.$$
We claim that there is a group structure on $E(R)$ which was proved a long time ago by Pascal. The group structure on $P(k)$ is defined by
$$-(x, y) = (x, -y) \quad \text{and} \quad P + Q + R = 0$$
for every $(x, y) \in \mathbb{P}^2(R)$ and every collinear $P, Q, R \in \mathbb{P}^2(k)$. Checking associativity amounts to some classical geometry. Note that E is an example of an algebraic group which is abelian, but it is not affine.

17.1. **Going between algebraic groups and Lie algebras.** If $f : G \to G'$ is a group homomorphism we naturally get a homomorphism of Lie algebras

$$D(f) : \mathrm{Lie}(G) \to \mathrm{Lie}(G').$$

Thus, Lie algebras are functorially attached to algebraic groups.

Example 17.2. We have an inclusion map $O(n) \to \mathrm{GL}(n)$ which gives us an inclusion of Lie algberas

$$\{\text{skew-symmetric matrices}\} \hookrightarrow \mathrm{Mat}(n, k).$$

A fundamental question is, how much information does this functor remember? In the discrete case, the answer is obviously *nothing*. Furthermore, the Lie algebras of $O(n)$ and $SO(n)$ are the same. This can be attributed to the fact that $O(n)$ is disconnected.

Definition 17.3. An algebraic group G is said to be *connected* if $\mathcal{O}(G)$ is a domain.

We will not prove the following proposition.

Proposition 17.4. *If $k = \mathbb{C}$ and $\mathcal{O}(G)$ is a finitely generated $\mathbb{C}$-algebra, then G is connected if and only if $G(\mathbb{C})$ is a connected topological space when viewed as a subspace of $\mathbb{C}^n$ with the Euclidean topology.*

Theorem 17.5. *If $\mathrm{char}(k) = 0$ and G, H are algebraic groups such that $\mathcal{O}(G), \mathcal{O}(H)$ are finitely-generated and G is connected. Then:*

$$\mathrm{Hom}(G, H) \longrightarrow \mathrm{Hom}(\mathrm{Lie}(G), \mathrm{Lie}(H))$$

is injective.

We will briefly sketch the proof when $k = \mathbb{C}$. We have a continuous map of topological spaces

$$\exp : \mathrm{Lie}(G) \longrightarrow G(\mathbb{C})$$

which is functorial. Another property of this exponential map that we have not defined is that exp is a local homeomorphism from an open neighbourhood of 0 to an open neighbourhood e. Then, one can show that if $D(f_1) = D(f_2)$ we have that

$$\{g \in G(\mathbb{C}) : f_1(g) = f_2(g)\}$$

is both open and closed.

18. Lecture 18

Let G be an algebraic group such that $\mathcal{O}(G)$ is a finitely generated $\mathbb{C}$-algebra. Recall that we want to construct a map

$$\exp : \mathrm{Lie}(G) \longrightarrow G(\mathbb{C})$$

with certain properties. In the case that $G \subseteq \mathrm{GL}(n)$ we can define $\exp(A) = \sum_n \frac{A^n}{n!}$, but we need something functorial and would therefore like to give an intrinsic construction.

Lemma 18.1. *For every $v \in \mathrm{Lie}(G)$, there exists a unique holomorphic homomorphism:*

$$\varphi_v : \mathbb{C} \to G(\mathbb{C})$$

such that $(d\varphi_v)_0(1) = v$.

Example 18.2. If $G = \mathrm{GL}(n)$ and $v = A \in M_n(\mathbb{C})$, we can define

$$\varphi_A(t) = \exp(tA) = \sum_n \frac{t^n A^n}{n!}.$$

The exponential map can be constructed as $\exp(v) = \varphi_v(1)$. Of course, there are several things we have to define for the lemma to make sense. For instance, what it means to be a holomorphic homomorphism in this context.

18.1. **Analytic structure.** Let more generally X be an affine variety of $\mathbb{C}$. We then define a topology on $X(\mathbb{C})$ as the weakest one such that for any $f \in \mathcal{O}(X)$,

$$f : X(\mathbb{C}) \to \mathbb{C}$$

is continuous.

Assume that X is of finite type and therefore there exists a surjection

$$\mathbb{C}[x_1, \dots, x_n] \longrightarrow \mathcal{O}(X)$$

corresponding to an embedding

$$X \longrightarrow \mathbb{A}^n_{\mathbb{C}}.$$

Definition 18.3. Let $U \subseteq X(\mathbb{C})$ be open. A function $h : U \to \mathbb{C}$ is *holomorphic* if and only if for any $a \in U$, h is the restriction of a holomorphic function on an open neighbourhood of a in $\mathbb{C}^n$.

Lemma 18.4. *This definition is independent of the embedding.*

Example 18.5. In general, an affine variety need not be a manifold. For instance, we can consider

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{y^2 - x^3}\right)$$

which has a cusp at the origin. However, it has a weaker structure of an analytic space. The failure to be a manifold is witnessed by the fact that

$$\dim(T_{X,(0,0)}) = 2$$

is larger than expected.

Definition 18.6. A function between analytic spaces is said to be *holomorphic* if the pullback of holomorphic functions are holomorphic.

Theorem 18.7. *If G is an algebraic group, then $G(\mathbb{C})$ is a holomorphic manifold.*

18.2. **Bringing ideas from ODEs.** To finish off the construction of φ_v, recall that any $v \in \mathrm{Lie}(G)$ determines a left-invariant vector field on G. We can define φ_v as the integral curve associated to the vector field and the existence and uniqueness theorem gives us a global existence, since the vector field is left-invariant. It follows from the uniqueness part of the theorem that $\varphi_v(t + t') = \varphi_v(t) \cdot \varphi_v(t')$. Then, we define

$$\exp(v) = \varphi_v(1)$$

and remark that

$$d\exp_0 = \mathrm{id}.$$

Thus, by the inverse function theorem, exp is a local isomorphism.